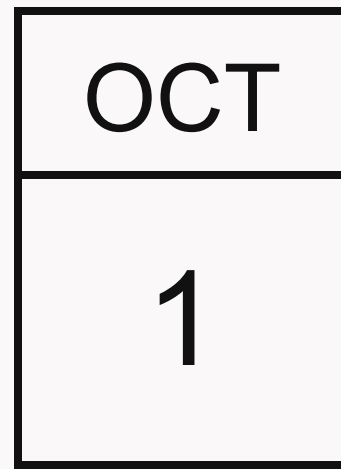


Name: _____

Date: _____



The Model T reaches its first customer

A first customer helped move the Model T from a factory idea into family life.

Have you ever seen a car and wondered how many small parts help it move? On October 1, 1908, a Ford Model T was shipped to its first customer. The car was not a magic machine; it was a careful collection of metal, glass, rubber, and moving pieces. The delivery connected Ford's Highland Park work with a real family waiting to use the car on public roads. The Model T mattered because Ford worked to make car travel more affordable for many families. That change did not happen in one afternoon, but one first customer made the date memorable. The first customer did not see every future change that the car would bring. Still, one Model T leaving the factory showed how an invention can move from a plan on paper into an ordinary person's life. That is why this small October delivery is worth remembering. The car's arrival was one small moment in the long history of travel. It showed how a new machine could leave a factory and become part of family life. The Model T story is therefore about more than a car: it is about an invention becoming useful to people outside the workshop.

Name: _____

Date: _____

OCT

Level

1

A Ford Model T has 4 tires. If the factory produces 3 cars today, how many tires must be installed in total?

Name: _____

Date: _____

Level

Q2

The Ford factory can assemble 12 cars in 1 hour. If workers continue for 8 hours, how many cars can they assemble in total?

Name: _____

Date: _____

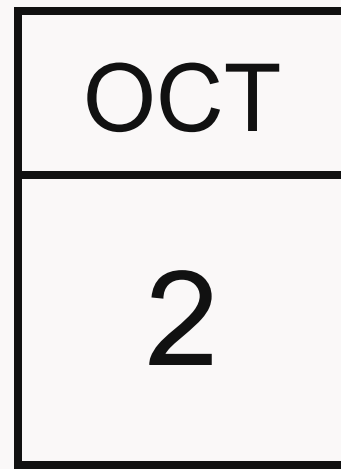
Level

3

The first Model T was sold in 1908 for \$850. If this year is 2026, how many years ago was it released?
If Xiaoming saves \$50 each month, how many months must he save to afford the car's original price?

Name: _____

Date: _____



Peanuts appears in newspapers

A few newspaper panels gave Peanuts room to make ordinary school days funny.

What can a comic strip show with only a few boxes and some drawings? On October 2, 1950, Charles Schulz's Peanuts first appeared in newspapers. Readers met characters who could make an ordinary school day feel funny, surprising, or a little awkward. Schulz drew the first strip for a newspaper page, where a few panels had to carry the action and the joke. A comic strip has to choose its words carefully because each panel has limited space. A small picture can show a big feeling, and a short line can make a reader laugh. Schulz kept drawing the characters after that first appearance, and readers returned to see what happened next. A newspaper comic may look small, but its characters can stay with people for many years. Peanuts began with a few panels and became part of daily reading. The first newspaper appearance was brief, but the characters returned again and again. Their ordinary problems helped readers recognize pieces of their own lives. Each new strip gave Schulz another chance to show a small feeling, and those small feelings helped the comic speak to many readers.

Name: _____

Date: _____

OCT

Level

1

A Peanuts comic strip has 4 panels, and each panel has 2 dogs. How many dogs are in the comic strip altogether?

Name: _____

Date: _____

Level

Q2

An artist draws 6 comic panels each day. If the artist draws for 3 weeks, with 7 days in each week, how many panels are drawn altogether?

Name: _____

Date: _____

Level

3

The Peanuts comic strip ran from 1950 to 2000.
How many years did it run? If it published an
average of 365 strips each year, about how many
strips were published in 50 years?

Name: _____

Date: _____



Transistor inventors receive a patent

A tiny transistor helped engineers imagine smaller electronic devices.

Have you ever opened a small electronic device and wondered what makes it work? On October 3, 1950, the inventors of the transistor received a patent. A transistor is a tiny part that can control an electric signal. The inventors developed the transistor at Bell Telephone Laboratories, where engineers needed a smaller way to control signals. The transistor helped lead to smaller and more useful electronics. A radio, computer, or phone may contain many parts that are hard to see, yet each one can have an important job. The patent did not make every electronic problem disappear. It recorded an idea that other engineers could study and improve. A tiny component may be easy to overlook, yet the transistor became an important part of the story of modern electronics. The patent recorded a useful idea at one point in time. Later inventors built on that idea, helping electronic devices become smaller and more powerful. The transistor's story shows how a part that is too small to notice can still help change the way people communicate and work.

Name: _____

Date: _____

OCT

Level

1

An engineer has 12 small transistors and accidentally breaks 4. How many transistors are still usable?

Name: _____

Date: _____

Level

Q2

A pocket radio needs 6 transistors. If a factory has 90 transistors, how many pocket radios can it assemble at most?

Name: _____

Date: _____

Level

3

A transistor patent was obtained in 1950. If a transistor cost \$8 each then and costs \$0.50 each today, how many dollars are saved on 100 transistors today?

Name: _____

Date: _____



Sputnik beeps from space

A few beeps from Sputnik showed that a human-made object was circling Earth.

If you heard a tiny beep from the sky, what would you guess was making it? On October 4, 1957, Sputnik 1 became the first artificial satellite to orbit Earth. It was a small metal spacecraft with antennas that sent beeps back to ground stations. Sputnik traveled around Earth after Soviet engineers launched it into orbit, while radio stations on the ground listened for its signal. People on Earth could listen for the signal even though Sputnik was far above them. The beeps showed that an object made by people was traveling around the planet. Sputnik did not carry people, but it changed what people believed was possible above Earth. Its signal crossed the distance between the satellite and listeners on the ground. A few beeps became a clear sign that a new space age had begun. The beeps were simple, yet they carried across space. People on the ground heard evidence that a human-made object was circling Earth. Space looked empty, but the signal proved that careful planning had sent a real object through it and brought information back home.

Name: _____

Date: _____

OCT

Level

1

Sputnik 1 had 4 metal antennas. How many antennas would 2 identical satellites have?

Name: _____

Date: _____

Level

Q2

Sputnik 1 took about 98 minutes to orbit Earth once. How many minutes would 2 orbits take? How many hours and minutes is that?

Name: _____

Date: _____

Level

3

Sputnik 1 weighed 83 kilograms. If a spacecraft can carry 500 kilograms, how many kilograms remain after loading one satellite? How many of these satellites can it carry at most?

Name: _____

Date: _____



World Teachers' Day begins

Teachers help turn difficult ideas into skills children can use.

Who has helped you learn something that once seemed difficult? World Teachers' Day is celebrated every October 5. UNESCO says the observance has been held since 1994, giving people a day to notice the work of teachers. The observance connects schools and communities around the world, where students and teachers share the work of learning. A teacher may explain the same idea in several ways until a student finds a way in. Good teaching is often made from many small acts: asking, listening, showing, and trying again. Teachers' Day gives families and students a chance to say thank you, but the work of teaching continues every day. A lesson may be remembered because a teacher explained it patiently or encouraged a child to try once more. A teacher's work may not make a loud sound like a rocket. Its effects can last quietly in the questions, skills, and confidence students carry forward. The history of a classroom is made from moments like these, when someone learns a new idea and later uses it to help another person.

Name: _____

Date: _____

OCT

Level

1

A teacher prepares 20 red pens and uses 8 for grading. How many red pens remain?

Name: _____

Date: _____

Level

Q2

A school has 24 classes with 2 homeroom teachers per class. How many homeroom teachers are there? If the principal gives each teacher 3 special pens, how many pens are needed?

Name: _____

Date: _____

Level

3

World Teachers' Day was established in 1994. By 2026, how many anniversaries have been celebrated? If a teacher taught 30 students each year from 1994, how many students did she teach in 32 years?

Name: _____

Date: _____



Moulin Rouge opens

A bright windmill helped visitors find a famous Paris theater.

What would you notice first if a theater had a giant windmill outside? The Moulin Rouge opened in Paris on October 6, 1889. Its famous red windmill became a sign that people could recognize from far away. The theater opened in Paris with its windmill placed high above the building, making the entrance easy for visitors to spot. A theater is more than a building with seats. It is also a place where lights, music, costumes, dancers, and an audience meet for one show. The windmill helped people recognize the Moulin Rouge from a distance. Inside, performers and musicians made the theater lively. The building became famous not only because of its shows, but also because its bold sign stood out in Paris. The bright windmill became a symbol of the theater. It helped turn a building in Paris into a place recognized by people far beyond its neighborhood. A theater can become part of a city's memory when its building, performances, and visitors keep returning to the same lively place.

Name: _____

Date: _____

OCT

Level

1

The Moulin Rouge has a windmill with 4 blades.
How many blades do 3 windmills have?

Name: _____

Date: _____

Level

Q2

A theater has 15 seats in each row and 8 rows.
How many people can the theater seat?

Name: _____

Date: _____

Level

3

When the Moulin Rouge opened in 1889, a ticket cost 5 francs. If 450 people attend tonight, what is the ticket income? If performers' wages cost 1,200 francs, how much profit remains?

Name: _____

Date: _____



The barcode patent is granted

A striped pattern gave checkout machines information to read.

Why might a cashier prefer a quick scan to typing every price? On October 7, 1952, Norman Woodland and Bernard Silver received a patent for a barcode design. Their idea became an early step toward faster store checkouts. Woodland and Silver designed a pattern that a machine could read, turning printed lines into information for a store system. A barcode is a pattern that a scanner can read. The pattern itself is simple to look at, but it gives a machine a way to find information quickly. The early patent was only one step in a long history of store technology. Later designs made scanning quicker and more common. The black-and-white pattern is simple, but it helps a computer find the right product information. A pattern made for machines became part of everyday shopping. The patent was an early step toward the quick checkout lines people know today. The barcode became useful because people agreed that a shared pattern could help machines and humans pass information to each other in everyday life.

Name: _____

Date: _____

OCT

Level

1

A barcode has 8 black lines and 7 white lines. How many lines are on the barcode?

Name: _____

Date: _____

Level

Q2

Entering a price by hand takes 12 seconds, while scanning takes 2 seconds. How many seconds are saved per item? How many seconds are saved for 10 items?

Name: _____

Date: _____

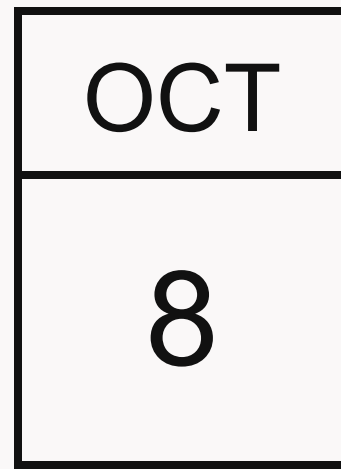
Level

3

A supermarket serves 120 customers in one hour. Each customer buys an average of 15 items. How many barcodes are scanned in one hour?

Name: _____

Date: _____



Cephalopod Awareness Days begin

What could you explore if you had eight arms and a puzzle to solve?

What could you explore if you had eight arms and a puzzle to solve? NOAA's Cephalopod Awareness Days begin on October 8 and celebrate fascinating ocean animals. NOAA explains that octopuses have eight arms and no tentacles. Their arms have suckers that help them sense the world around them. An octopus can use its arms to explore and manipulate objects, much like a child testing how a puzzle piece fits. NOAA educators describe octopuses as intelligent animals and explain that one octopus named Pi learned to open jars and work with food puzzles. That does not mean an octopus thinks exactly like a person. It means animals can solve problems in ways that fit their bodies and habitats. Octopuses can also change color, crawl across the sea floor, and squeeze into small spaces. On October 8, the observance gives us a reason to notice these unusual animals and ask how scientists learn from them. The next time you see a puzzle, imagine having eight helpful arms ready to investigate it.

Name: _____

Date: _____

OCT

Level

1

An octopus has 8 arms. How many arms do 2 octopuses have?

Name: _____

Date: _____

Level

Q2

An octopus has 3 hearts and 9 brains. How many hearts and how many brains do 5 octopuses have?

Name: _____

Date: _____

Level

3

A giant Pacific octopus can weigh 50 kilograms, while a young octopus weighs 200 grams. How many times heavier must the young octopus become to reach 50 kilograms? Use 1 kilogram = 1,000 grams.

Name: _____

Date: _____



The Universal Postal Union is founded

Shared postal rules helped messages cross borders.

How can a letter travel across a country when people speak different languages? The Universal Postal Union was founded on October 9, 1874. It helped countries work together so international mail could follow shared rules. The agreement was made by countries working through an international postal organization so mail could cross national borders. A letter still needs an address, a carrier, and a journey, but cooperation can make the journey easier to organize. World Post Day remembers this kind of connection. The agreement did not make every letter travel quickly, but it gave postal services a way to cooperate. A message could cross a border more easily when workers shared rules. The organization helped connect people who lived far apart. A letter may be small, but its journey crosses real borders. Shared postal rules helped a message travel between people who lived in different countries. Postal history is also human history: every organized route exists because someone wanted news, a greeting, or an important message to arrive across the world.

Name: _____

Date: _____

OCT

Level

1

Ming mails 7 letters and Hua mails 9 letters. How many letters do they mail altogether?

Name: _____

Date: _____

Level

Q2

A mail carrier delivers 8 boxes each day, with 25 letters in each box. How many letters are delivered each day?

Name: _____

Date: _____

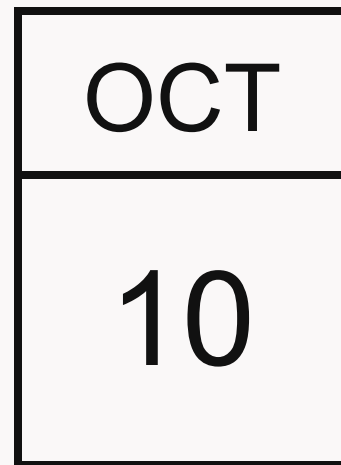
Level

3

When the Universal Postal Union was founded in 1874, an international letter cost 15 cents. If it costs \$1.50, or 150 cents, today, how many times the old postage is the current postage?

Name: _____

Date: _____



World Mental Health Day

A short conversation can be an important first step toward mental-health support.

What helps you feel ready to talk when a hard day is bothering you? World Mental Health Day has been observed on October 10 since 1992. The day reminds people that feelings and thoughts deserve care, just as bodies do. The observance brings attention to care in homes, schools, and health services, where people can notice feelings and seek support. Care can begin with a quiet conversation, a walk, a trusted adult, or a plan to get more help. No single activity fixes every problem, but asking for support can be a strong first step. The day also reminds people to listen without judging. Someone may look cheerful while carrying a difficult feeling. Kind words and professional help can both matter, and asking for support is a responsible choice rather than a weakness. A day about mental health makes room for quiet experiences that are easy to miss. Care begins when people take feelings seriously and help each other find support. Taking care of mental health includes noticing when a friend needs kindness and knowing that trained helpers can provide more support when needed.

Name: _____

Date: _____

OCT

Level

1

Gao sleeps for 9 hours and Ming sleeps for 8 hours. How many hours do they sleep altogether?

Name: _____

Date: _____

Level

Q2

A doctor recommends walking for 30 minutes each day. If Mei walks for 2 weeks, or 14 days, how many minutes does she walk? How many hours is that?

Name: _____

Date: _____

Level

3

The daily goal is 10,000 steps. Qiang walks 3,850 steps in the morning and 4,200 in the afternoon. How many more steps are needed to reach the goal?

Name: _____

Date: _____



The International Day of the Girl

Fair opportunities can help more girls learn, choose, and create.

What opportunity would you want every child to have? October 11 highlights girls' rights and opportunities to learn and grow. A school, a library, or a science club can open a door when children are welcomed and supported. The day is observed in communities that are working to make schools, safety, and opportunities fairer for girls. The date is not about one famous object; it is about people and the choices communities make. Fair chances let more children practice skills, ask questions, and imagine future work. The date asks people to notice barriers that can keep girls from learning or making choices about their futures. When schools and communities offer fair chances, more children can discover what they enjoy and what they can do. The day asks communities to remove unfair barriers. When girls receive chances to learn, their ideas and abilities can shape the future in many different ways. The opportunity to learn can change a child's choices, and fair access allows more girls to become the people who solve tomorrow's problems.

Name: _____

Date: _____

OCT

Level

1

A library buys 15 storybooks and lends 7. How many books remain?

Name: _____

Date: _____

Level

Q2

A charity puts 45 books in each box and donates 6 boxes. How many books are donated?

Name: _____

Date: _____

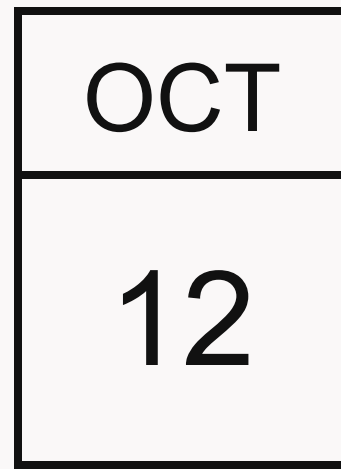
Level

3

A foundation plans to support 500 girls. Tuition and books cost \$120 per girl each year. How much money must the foundation raise each year?

Name: _____

Date: _____



Many astronauts share one spacecraft

A crowded spacecraft made teamwork part of the mission.

How would a crew share one small spacecraft during a long trip? Voskhod 1 launched on October 12, 1964, carrying the first crew of more than one person without spacesuits. The astronauts had to work inside a tight spacecraft. The spacecraft carried its crew in orbit, where the astronauts had to complete work inside a cabin with very little room. Space travel depends on careful planning because food, air, tools, and time are limited. A crew must know what to do next and how long each task should take. The flight was crowded because the crew had to work without spacesuits inside the spacecraft. The astronauts shared a small space and followed a careful plan. Voskhod 1 showed that spaceflight could involve a crew working together. The crew had to share space, tools, and time. Their flight showed that space exploration could involve people working together inside one spacecraft. The astronauts' shared cabin made their teamwork visible. Every person had a role in keeping the mission organized during the flight far from home.

Name: _____

Date: _____

OCT

Level

1

Voskhod 1 carried 3 astronauts. How many astronauts would 2 identical spacecraft carry?

Name: _____

Date: _____

Level

Q2

The spacecraft operated for 24 hours. If astronauts conducted one experiment every 3 hours, how many experiments did they conduct?

Name: _____

Date: _____

Level

3

The spacecraft traveled about 670,000 kilometers in 16 orbits around Earth. What was the average distance per orbit?

Name: _____

Date: _____



Disaster risk gets its own day

A plan made before danger can help a community respond.

What would you put in a kit to help your class during an emergency?
October 13 is the International Day for Disaster Risk Reduction. The day encourages people to notice risks and prepare before a dangerous event happens. The international observance helps families, schools, and local officials compare risks and practice a response before a disaster. Preparation can be simple: know where to go, keep supplies together, listen to trusted instructions, and practice a plan. A plan cannot stop every emergency, but it can help people respond. Preparation begins before people see smoke, water, wind, or another danger. A family can learn an alarm plan, and a school can practice leaving safely. The October day turns attention toward reducing risk before an emergency starts. The best emergency plan is made before people need it. Knowing a safe route and listening to good instructions can help a community respond calmly. Risk reduction is practical history because communities learn from earlier disasters and use that knowledge to make future decisions safer.

Name: _____

Date: _____

OCT

Level

1

An emergency kit has 5 water bottles, 4 food packs, and 2 flashlights. How many items are in the kit?

Name: _____

Date: _____

Level

Q2

For a school disaster drill, each class needs 40 food packs. If there are 18 classes, how many packs are needed?

Name: _____

Date: _____

Level

3

An emergency food pack expires after 3 years. If it was bought in October 2024, when does it expire?
If each of 50 students uses 2 water bottles per day, how many days will 1,200 bottles last?

Name: _____

Date: _____



An airplane breaks the sound barrier

One difficult flight gave engineers new information about speed.

What might an airplane pilot hear when the plane flies faster than sound? On October 14, 1947, Chuck Yeager flew faster than sound in the Bell X-1 aircraft. The flight tested an idea that had seemed very difficult and dangerous. The Bell X-1 was released from a larger aircraft and used its engine to test what happened as it moved faster through the air. The airplane had to be designed for unusual speed and pressure. Scientists and engineers studied measurements carefully because a tiny error could matter in the air. The Bell X-1 looked different from an ordinary airplane because it was built for a special test. Yeager's flight gave engineers new information about high-speed travel. The sound barrier was not a wall, but it was a difficult problem that pilots had to understand. The flight gave engineers information they could use for later aircraft. A difficult test became part of the step-by-step history of faster travel. The sound barrier flight added a new piece of evidence to aviation history, where each test helps explain what an aircraft can do.

Name: _____

Date: _____

OCT

Level

1

The Bell X-1 airplane flown by Yeager was 9 meters long. How long would 2 such airplanes be altogether?

Name: _____

Date: _____

Level

Q2

The airplane reached 1,066 kilometers per hour when it broke the sound barrier. How far would it travel in 2 hours at that speed?

Name: _____

Date: _____

Level

3

Sound travels through air at about 340 meters per second. How far does an airplane travel at that speed in 15 seconds? How many kilometers and meters is that?

Name: _____

Date: _____



Clean hands get a special day

Soap, water, and a small routine can protect a classroom.

How long do you spend washing your hands before eating? The first Global Handwashing Day was held on October 15, 2008. It taught that soap and water can help remove germs and prevent illness. The public-health event takes place in schools and communities, where people practice cleaning their hands with soap and water. A useful habit is made from a small action repeated at the right time. Rubbing hands, rinsing, and drying may feel ordinary, but those steps can protect a whole classroom. Handwashing is easy to overlook because it takes only a short time. The day gives schools a reason to practice the steps together. Soap, water, and a little patience can turn a simple routine into useful protection. Clean hands are a small defense against illness. A shared reminder can help many people repeat the habit at home, at school, and in public places. The lesson travels farther than one classroom because children can carry the habit to their families and make clean hands part of daily life.

Name: _____

Date: _____

OCT

Level

1

Proper handwashing takes 20 seconds. Ming washes for 20 seconds and Hua washes for 15 seconds. How many seconds do they wash altogether?

Name: _____

Date: _____

Level

Q2

A class has 30 students, and each student washes hands 4 times at school each day. How many handwashes are there in one day?

Name: _____

Date: _____

Level

3

A bar of soap can wash hands about 150 times. If a family of 4 washes 5 times per person each day, how many days will one bar last?

Name: _____

Date: _____



Disney Brothers Cartoon Studio begins

A written agreement helped two brothers begin an animation studio.

What could two brothers make with drawings, patience, and a small studio? On October 16, 1923, Walt and Roy Disney signed a contract that began their cartoon studio story. Their work grew from drawings and teamwork. The brothers worked in a small early studio and used a written agreement to begin building a business around animated drawings. Animation takes many drawings because a moving picture is made from images shown in order. Artists repeat a motion carefully so a character appears to walk, wave, or smile. A contract is not the same as a finished cartoon, but it marked the beginning of a studio that would grow. Walt and Roy Disney still needed drawings, stories, workers, and many experiments. Their first agreement was one early step in that work. The contract was only a beginning. Drawings, stories, music, and teamwork would be needed before an idea on paper could become a finished cartoon. A studio begins with people who choose to work together. The drawings matter, but so do the agreements that let the work continue.

Name: _____

Date: _____

OCT

Level

1

Walt Disney has 15 pencils, breaks 4, and receives 8 more. How many pencils does he have now?

Name: _____

Date: _____

Level

Q2

A 7-minute cartoon needs 10,000 drawings. If an animator draws 100 each day, how many days are needed?

Name: _____

Date: _____

Level

3

Disney was founded in 1923. By 2026, how many years old is it? If a major celebration is held every 5 years, how many complete celebrations have occurred?

Name: _____

Date: _____



Small actions can help communities

Communities can make help easier to reach when they listen and organize.

What small action could make a neighbor's day easier? October 17 is a day for learning how communities can work together to reduce poverty. The focus is on people, resources, and practical help rather than one single invention. The observance takes place in communities where people and organizations discuss practical ways to remove barriers and share resources. A community can share food, teach a skill, repair a home, or listen to a family. These actions do not make every challenge disappear, but they can make help easier to reach. The day asks people to look at poverty as a problem that affects real neighbors, not as a distant number. Communities can listen to what families need and share power, time, food, or knowledge. Small actions matter more when people organize them together. Reducing poverty takes more than one kind gesture. It depends on people listening, sharing resources, and changing systems that make daily life harder. A community becomes stronger when help is planned with the people who need it, instead of being designed without listening to them.

Name: _____

Date: _____

OCT

Level

1

Volunteers collect 25 bags of rice and 18 bags of flour. How many food bags are there altogether?

Name: _____

Date: _____

Level

Q2

A food bank receives 240 cans and divides them equally into 12 care boxes. How many cans go in each box?

Name: _____

Date: _____

Level

3

A community sale prepares 150 meals. Each sells for \$8 and costs \$3 to make. If all meals sell, how much money is raised after costs?

Name: _____

Date: _____



The BBC begins broadcasting

Radio brought the same voices and music into separate homes.

What would you choose to hear if you could turn a radio dial in 1922? The British Broadcasting Company began broadcasting on October 18, 1922. Radio let people hear music, news, and stories in their homes. The company began in Britain, sending scheduled programs from a station to listeners who owned radio receivers at home. A broadcast travels from a station to listeners who may be far away. The people speaking into a microphone cannot see every listener, so clear sound and careful timing matter. The first broadcasts were new to many listeners, who had to learn how to use this unfamiliar kind of home entertainment. A voice could travel through a box and reach a room far away. Broadcasting changed how people shared news and stories. The first broadcast introduced many listeners to a new way of sharing sound. Voices and music could now travel into homes that were far from the station. Broadcasting made a new kind of shared moment. People in separate homes could hear the same program and imagine that they were listening together.

Name: _____

Date: _____

OCT

Level

1

A radio plays 3 hours of music and 2 hours of news. How many hours does it play altogether?

Name: _____

Date: _____

Level

Q2

A radio station plays 4 story songs each hour. If it broadcasts for 14 hours, how many story songs does it play?

Name: _____

Date: _____

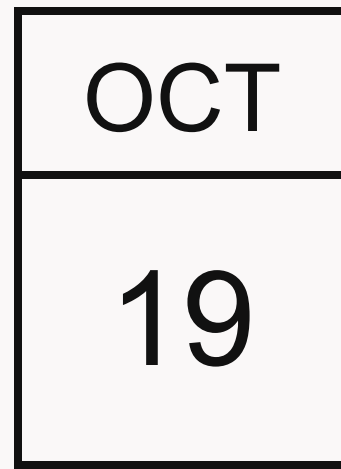
Level

3

A radio program lasts 45 minutes and airs once each week. Over 52 weeks, how many minutes does it air? How many hours is that?

Name: _____

Date: _____



Repair instead of replace

Repairing an object can save materials and teach a useful skill.

What object would you fix instead of throwing away on an October repair day? Repair events encourage people to fix useful objects rather than replace them. A loose button, a broken lamp, or a bicycle chain may need patience more than a new purchase. Repairers work in homes, shops, and community spaces, where they inspect an object, find the broken part, and make it useful again. Repairing also teaches people to look closely. First find the problem, then choose a safe tool, make one change, and test the object again. Repairing an object can save materials and teach a useful skill. It can also bring neighbors together around a table. A repaired lamp or bicycle may not look brand new, but it can keep doing its job instead of becoming waste. A repaired object keeps its earlier story while becoming useful again. Repair work can save materials and give people a reason to learn from one another. Repair culture values care for objects and care for resources. It asks what can be saved before a useful thing is thrown away.

Name: _____

Date: _____

OCT

Level

1

A repair station receives 14 broken toy cars and fixes 9. How many are still broken?

Name: _____

Date: _____

Level

Q2

A new lamp costs \$45, while parts to repair an old lamp cost \$8. How much is saved by repairing one lamp? How much is saved on 5 lamps?

Name: _____

Date: _____

Level

3

Repairing one bicycle takes 45 minutes. If a repairer works for 5 hours, or 300 minutes, how many bicycles can be repaired at most?

Name: _____

Date: _____



The first World Statistics Day

Careful counting helps people understand what numbers really show.

How can numbers help us notice what is happening in a group? World Statistics Day was first celebrated worldwide on October 20, 2010. Statistics are numbers collected and organized so people can look for patterns. The international day uses careful counting and shared definitions so people in different places can compare information fairly. A number alone does not answer every question. We also need to know what was counted, when it was counted, and whether the way of counting was fair. A statistic needs a clear question behind it. If people count different things or use different rules, the numbers can mislead. World Statistics Day highlights the careful work needed to collect, check, and explain information. Reliable statistics help people describe change without guessing. The counting method matters because a number is only useful when everyone understands what it represents. Statistics can support good decisions only when the collection is honest, the categories are clear, and the people reading the result understand its limits for everyone to use.

Name: _____

Date: _____

OCT

Level

1

A chart records 12 people who like apples and 8 who like bananas. How many people are in the two groups altogether?

Name: _____

Date: _____

Level

Q2

Four classes record 120, 135, 110, and 155 books read. How many books do the classes read altogether?

Name: _____

Date: _____

Level

3

Ming records 8,000, 9,500, 10,000, 7,500, and 10,000 steps over 5 days. What is the average number of steps per day?

Name: _____

Date: _____



Edison tests a long-lasting lamp

A lamp became useful when its light could last.

How long would you want a lamp to stay lit during a power cut? Thomas Edison demonstrated a practical incandescent lamp that stayed lit for many hours in October 1879. A lamp that lasted longer was useful because people did not need to replace it quickly. Edison tested lamps in his New Jersey laboratory, changing materials and designs until the light could last long enough to be useful. Early inventors tested materials again and again. A bright idea was only useful when the lamp could be made, turned on, and kept working safely. Edison and other inventors tested many materials before a lamp could glow for a useful length of time. A light that stayed on was more helpful than one that burned out quickly. The demonstration showed why testing matters in an invention. A lamp that stayed lit was valuable because people needed light they could trust. Testing materials carefully helped turn a bright idea into a practical tool. A reliable lamp changed an ordinary evening. It gave people a steadier way to see, read, and work after the sun went down.

Name: _____

Date: _____

OCT

Level

1

The first light bulb stayed lit for 13 hours. If 2 bulbs take turns, how many hours can they stay lit altogether?

Name: _____

Date: _____

Level

Q2

One LED bulb uses 10 watts. If 6 bulbs are installed in a living room, how many watts do they use? If they run for 5 hours, how many watt-hours do they use?

Name: _____

Date: _____

Level

3

A modern energy-saving bulb lasts 15,000 hours. If it is lit for 10 hours each day, about how many days does it last? About how many years is that, using 365 days per year?

Name: _____

Date: _____



A parachute opens over Paris

A parachute turned a dangerous fall into a slower descent.

What would you notice if a parachute opened above a busy city? On October 22, 1797, André-Jacques Garnerin made the first recorded parachute jump from a balloon. He tested whether a wide canopy could slow a fall. Garnerin rose above a crowd in Paris and opened a fabric canopy, which slowed his fall as he descended toward the ground. A parachute works because air pushes against the canopy. The jump was risky, so the design had to open, spread, and carry a person safely toward the ground. Garnerin's jump was a test as well as a spectacle. The parachute had to open and slow his fall, and people on the ground watched closely. The event helped show that a canopy could give a person a safer way to descend through the air. A parachute changes a dangerous fall into a slower descent. Garnerin's jump gave people a memorable test of an idea that had once seemed impossible. The jump became a chapter in the history of flight, when people were testing ways to move through the air without an engine.

Name: _____

Date: _____

OCT

Level

1

Gagarin rises to 1,000 meters in a balloon and parachutes down. If he descends 300 meters, how far above the ground is he?

Name: _____

Date: _____

Level

Q2

A parachutist lands safely after 120 seconds. How many minutes is that? If the parachutist jumps 3 times, how many minutes are spent in the air?

Name: _____

Date: _____

Level

3

A parachutist jumps from 1,000 meters. After the parachute opens, the average descent speed is 5 meters per second. How many seconds does the descent take? How many minutes and seconds is that?

Name: _____

Date: _____



Pelé is born

A football career grows from skill, practice, teamwork, and choices.

What can a player learn by passing the ball instead of keeping it? Pelé was born on October 23, 1940, and became one of football's most famous players. His career included goals, matches, practice, and teamwork. Pelé played on football fields in Brazil and around the world, using practice and quick decisions to help his team attack. A great player does not work alone. Teammates watch the field, move into space, and make choices quickly while the game keeps changing. Pelé's skill made him famous, but football is played with teammates and opponents on the same field. A player must practice, notice openings, and make quick choices. His life is remembered through the goals and matches that shaped his career. Pelé's career joined skill with practice and teamwork. His goals are part of the record, but the people on the field helped make each match possible. A match is remembered in goals, but a career also contains practice, travel, teamwork, and choices made when no camera is watching over many seasons.

Name: _____

Date: _____

OCT

Level

1

Pelé scores 2 goals in the first half and 1 in the second half. How many goals does he score altogether?

Name: _____

Date: _____

Level

Q2

A soccer game lasts 90 minutes. If Pelé plays 10 full games, how many minutes does he play?

Name: _____

Date: _____

Level

3

Pelé scores 1,280 goals over a 20-year active career. How many goals per year is that on average? If he plays 32 games per year, how many goals per game is that on average?

Name: _____

Date: _____



Annie Taylor rides over Niagara Falls

A dangerous Niagara attempt became a story about testing a protective design.

Would you ride inside a barrel over a waterfall? On October 24, 1901, teacher Annie Edson Taylor survived going over Niagara Falls in a barrel. She attempted the daring challenge after planning and preparing for it. Taylor entered the Niagara River in a barrel and went over the waterfall, making the event a dangerous test of a protective design. The story is exciting, but it also reminds us that dangerous stunts should never be copied. Water, height, and speed can make a small mistake very serious. Taylor's barrel was designed for the dangerous trip, and her survival made the event famous. The story is not a safe experiment for children to copy. It is a reminder that courage and careful preparation do not remove every danger. Taylor's survival made her attempt famous, but the danger remained real. The event belongs to history, not to a list of stunts children should copy. The Niagara story is unusual because one person made a dangerous decision and survived. It is important to tell the story without turning danger into a game.

Name: _____

Date: _____

OCT

Level

1

Annie Taylor is 63 when she attempts her barrel challenge. How old is she 5 years later?

Name: _____

Date: _____

Level

Q2

Niagara Falls is about 51 meters high. If one building floor is about 3 meters high, how many floors high is the waterfall?

Name: _____

Date: _____

Level

3

A barrel falls 51 meters from the top of the falls to the water in about 3 seconds. What is its average speed in meters per second? What is that speed in kilometers per hour?

Name: _____

Date: _____



Picasso is born

Changing shapes and viewpoints helped Picasso explore new kinds of art.

What shapes and colors would you choose for a painting? Pablo Picasso was born on October 25, 1881, in Málaga, Spain. His art changed over a long life, and many people still study his choices of shape, color, and viewpoint. Picasso worked in studios in Spain and France, changing his lines, shapes, and viewpoints as he explored new kinds of art. An artist may show one object from more than one angle. Looking carefully can help us notice that a picture is not only about copying what the eye sees. Picasso's long life gave him time to change his style and try different ways to show the world. A painting can ask viewers to look twice. Shapes, colors, and unusual viewpoints can make one subject feel new. Picasso's art did not stay in one style. His paintings show how an artist can keep exploring instead of using the same way to make every picture. An artist's changing work can invite viewers to ask why a picture looks different and what the artist wanted them to notice.

Name: _____

Date: _____

OCT

Level

1

A Picasso painting has 5 blue triangles and 6 red circles. How many shapes are in the painting?

Name: _____

Date: _____

Level

Q2

Picasso created about 1,800 oil paintings. If an artist made the same number over 30 years, how many would be made each year on average?

Name: _____

Date: _____

Level

3

Picasso was born in 1881 and died in 1973. How old was he? If he created professionally from age 15 to age 90, how many years was that?

Name: _____

Date: _____



The Erie Canal opens

A canal gave boats a new route through a difficult landscape.

How could a canal change the trip between two places? The Erie Canal opened on October 26, 1825. It linked Lake Erie with the Hudson River and created a new route for boats, people, and goods. The canal joined waterways in the United States, letting boats move between the Great Lakes and the Hudson River more directly. Before the canal, wagons had to cross rough roads and hills. A water route could carry heavy loads in a different way, although the trip still took planning and time. The canal did not erase every difficulty in travel, but it offered a new route for heavy goods. Boats could move between important waterways, helping towns along the route grow. The opening changed how people imagined travel through the region. The canal changed the movement of goods across the region. Its opening connected waterways and gave travelers a new route through a landscape that had been hard to cross. Routes change people's daily lives. A canal can affect where goods travel, where towns grow, and how long a journey takes.

Name: _____

Date: _____

OCT

Level

1

The Erie Canal has 8 cargo boats and 5 passenger boats. How many boats are on the canal?

Name: _____

Date: _____

Level

Q2

Before the Erie Canal, transporting goods by wagon took 20 days. After the canal opened, boats took 8 days. How many days were saved?

Name: _____

Date: _____

Level

3

The Erie Canal is about 584 kilometers long. At 8 kilometers per hour, how many hours does a boat need to travel the whole canal? How many days and hours is that?

Name: _____

Date: _____



New York's subway opens

A subway moved many passengers below a crowded city.

What would a train station sound like when a new subway line opens? New York City's first subway line opened on October 27, 1904. The subway gave riders a way to travel below busy streets. The first subway trains ran beneath New York City streets, using tracks and stations to move passengers through a busy city. A subway station needs more than a train. It needs tracks, signals, platforms, workers, and rules that help many passengers move safely. The subway line was a new kind of trip for New Yorkers. Trains could carry many people below streets that were already crowded above. A subway depends on shared rules because one safe journey requires many workers and passengers to cooperate. The subway carried people below streets that were already crowded. Its success depended on tracks, signals, workers, and passengers following shared rules. The subway's history is a story of shared space. A safe trip asks thousands of strangers to follow the same simple rules in a busy city for city riders today.

Name: _____

Date: _____

OCT

Level

1

A subway car has 18 passengers. At a station, 6 leave and 9 board. How many passengers are in the car now?

Name: _____

Date: _____

Level

Q2

A one-way New York subway ticket cost 5 cents in 1904. How many tickets can be bought with a 50-cent coin?

Name: _____

Date: _____

Level

3

A New York subway ride costs \$2.90 today and cost \$0.05 in 1904. How many times greater is today's fare? If someone rides twice a day for 22 workdays, what is the monthly cost?

Name: _____

Date: _____

OCT
28

The Statue of Liberty is dedicated

A torch above the harbor became a welcome for arriving travelers.

What might you see first when a statue stands at the entrance to a harbor? The Statue of Liberty was dedicated in New York Harbor on October 28, 1886. The statue became a landmark that many arriving visitors could see. The statue stands on an island in New York Harbor, where ships and visitors can see its torch above the water. A landmark can carry an idea as well as stand in a place. People may connect the statue with welcome, hope, and the journeys of families who arrived by sea. The statue was a gift from France and became a landmark in New York Harbor. People arriving by ship could see it before reaching the city. Its raised torch and welcoming position made it a powerful part of the harbor view. The statue became part of the harbor view for people arriving by ship. Its torch and welcoming position gave the landmark a meaning beyond its size. Landmarks collect stories from many visitors. The statue stayed in one harbor while the people looking at it came from many different places.

Name: _____

Date: _____

OCT

Level

1

The Statue of Liberty's crown has 7 rays. How many rays do 3 souvenir crowns have?

Name: _____

Date: _____

Level

Q2

Visitors climb 354 steps to the top. If Hua climbs 200 steps, how many more steps are needed?

Name: _____

Date: _____

Level

3

The Statue of Liberty weighs 225 metric tons, and 1 metric ton is 1,000 kilograms. How many kilograms is that? If a car weighs 1.5 metric tons, how many cars weigh the same as the statue?

Name: _____

Date: _____



The first ARPANET message begins with “LO”

A tiny network message marked the beginning of a much larger system.

What is the shortest message you could send to show that a connection works? On October 29, 1969, a UCLA team began sending a message toward the Stanford Research Institute through ARPANET. The system stopped after the letters LO. Engineers connected two computers in California and sent the first letters across the new network before the system stopped. The unfinished message was still useful because it showed that the connection had started. Later networks grew from many tests, repairs, and careful improvements. The system stopping after LO was not the end of the project. It showed that the first connection had begun, and engineers could study what went wrong before trying again. Early networks grew through tests, mistakes, and careful repairs. The letters LO marked a beginning even though the message stopped early. Engineers could use the failed attempt to repair the system and try again. The first network message was tiny, but later connections depended on beginnings like this one. A short test can lead to a much larger system.

Name: _____

Date: _____

OCT

Level

1

Scientists tried to send the word LOGIN, which has 5 letters, but the system crashed after sending LO, which has 2 letters. How many letters were not sent?

Name: _____

Date: _____

Level

Q2

A signal travels 7.5 times around Earth in 1 second.
If Earth's circumference is 40,000 kilometers, how
far does it travel in 1 second?

Name: _____

Date: _____

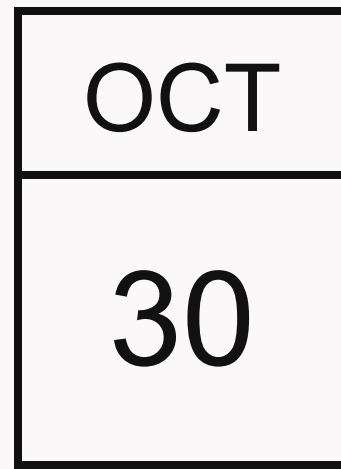
Level

3

A 5G network can send about 1,250,000,000 English letters each second. How many times the 2-letter 1969 message is that amount?

Name: _____

Date: _____



A radio drama sounds like news

A fictional radio emergency showed why listeners should check the facts.

How could a radio story sound so real that someone mistakes it for news? Orson Welles's radio adaptation of *The War of the Worlds* aired on October 30, 1938. Some listeners mistook parts of the fictional broadcast for real news. The radio drama was broadcast from a studio in New York, where actors performed a fictional emergency for listeners at home. Radio gives listeners only sound, so music, voices, and timing can create a strong picture in the mind. A careful listener should check the source before believing a surprising report. A radio story can feel immediate because listeners hear voices without seeing a stage. That is also why checking a source matters. Fiction can sound urgent, but a careful listener asks whether the report is a story or a real announcement. The broadcast showed how powerful sound can be when listeners cannot see the stage. It also showed why people should check whether an urgent report is fiction or news. The event is remembered because it shows both the strength and the limit of sound: a voice can create a picture, but it cannot replace checking facts.

Name: _____

Date: _____

OCT

Level

1

Six people are listening to a radio drama, and 5 neighbors join them. How many listeners are there?

Name: _____

Date: _____

Level

Q2

A radio drama lasts 60 minutes. The first 20 minutes are music, and the rest is the invasion story. How many minutes is the story? What fraction of the drama is it?

Name: _____

Date: _____

Level

3

About 6,000,000 people listen to a radio drama, and 1,200,000 panic because they think aliens have invaded. What percentage of listeners panic?

Name: _____

Date: _____



Halloween's older seasonal roots

Autumn traditions carry different meanings in different families.

What signs tell you that autumn is changing the world around you? Halloween is connected with older autumn traditions about the changing season and remembering the dead. Different communities have added their own foods, costumes, songs, and games. Autumn traditions developed in different communities, where families adapted older customs into the foods, stories, and activities they know today. A tradition can change as it travels from one family to another. The important part is to learn what a custom means to the people who practice it instead of assuming every family celebrates in the same way. Autumn traditions are not identical everywhere. One family may carve a pumpkin, another may share a special meal, and another may remember loved ones quietly. Learning the history of a tradition works best when we listen to the people who keep it. Autumn traditions carry different meanings for different families. Their variety is part of the history, because customs change as people share them across time and place. A tradition can preserve memories while still changing shape. Its history belongs to the people who explain, celebrate, and question it over time.

Name: _____

Date: _____

OCT

Level

1

Ming gets 18 candies for Halloween, eats 6, and receives 4 more. How many candies does he have now?

Name: _____

Date: _____

Level

Q2

A T. rex tooth is 30 centimeters long, and Ming's palm is 10 centimeters long. How many times longer is the tooth?

Name: _____

Date: _____

Level

3

A giant dinosaur weighs an estimated 60 metric tons, while an African elephant weighs about 5 metric tons. How many elephants weigh the same as the dinosaur? If it eats 400 kilograms of plants each day, how many metric tons does it eat in 30 days?

Answer Keys (Level 1)

Q1: 12 tires	Q2: 8 dogs
Q3: 8 transistors	Q4: 8 antennas
Q5: 12 pens	Q6: 12 blades
Q7: 15 lines	Q8: 16 arms
Q9: 16 letters	Q10: 17 hours
Q11: 8 books	Q12: 6 astronauts
Q13: 11 items	Q14: 18 meters
Q15: 35 seconds	Q16: 19 pencils
Q17: 43 bags	Q18: 5 hours
Q19: 5 cars	Q20: 20 people
Q21: 26 hours	Q22: 700 meters
Q23: 3 goals	Q24: 68 years old
Q25: 11 shapes	Q26: 13 boats
Q27: 21 passengers	Q28: 21 rays
Q29: 3 letters	Q30: 11 listeners
Q31: 16 candies	

Answer Keys (Level 2)

Q1: 96 cars	Q2: 126 panels
Q3: 15 radios	Q4: 196 minutes, or 3 hours 16 minutes
Q5: 48 teachers; 144 pens	Q6: 120 people
Q7: 10 seconds per item; 100 seconds for 10 items	Q8: 15 hearts; 45 brains
Q9: 200 letters	Q10: 420 minutes, or 7 hours
Q11: 270 books	Q12: 8 experiments
Q13: 720 food packs	Q14: 2,132 kilometers
Q15: 120 handwashes	Q16: 100 days
Q17: 20 cans	Q18: 56 songs
Q19: \$37 per lamp; \$185 for 5 lamps	Q20: 520 books
Q21: 60 watts; 300 watt-hours	Q22: 2 minutes; 6 minutes
Q23: 900 minutes	Q24: 17 floors
Q25: 60 paintings per year	Q26: 12 days
Q27: 10 tickets	Q28: 154 steps
Q29: 300,000 kilometers	Q30: 40 minutes; 2/3 of the drama
Q31: 3 times	

Answer Keys (Level 3)

Q1: 118 years; 17 months	Q2: 50 years; 18,250 strips
Q3: \$750	Q4: 417 kilograms; 6 satellites
Q5: 32 anniversaries; 960 students	Q6: 2,250 francs income; 1,050 francs profit
Q7: 1,800 barcodes	Q8: 250 times
Q9: 10 times	Q10: 1,950 steps
Q11: \$60,000	Q12: 41,875 kilometers
Q13: October 2027; 12 days	Q14: 5 kilometers 100 meters
Q15: 7.5 days	Q16: 103 years; 20 complete celebrations
Q17: \$750	Q18: 2,340 minutes, or 39 hours
Q19: 6 bicycles, with 30 minutes left	Q20: 9,000 steps
Q21: 1,500 days; about 4.11 years, or about 4 years	Q22: 200 seconds, or 3 minutes 20 seconds
Q23: 64 goals per year; 2 goals per game	Q24: 17 meters per second; 61.2 kilometers per hour
Q25: 92 years old; 75 years	Q26: 73 hours, or 3 days 1 hour
Q27: 58 times; \$127.60	Q28: 225,000 kilograms; 150 cars
Q29: 625,000,000 times	Q30: 20%
Q31: 12 elephants; 12 metric tons	